Nameer Iqbal Ansari

RTL Engineer | FPGA | RISC-V

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PROFESSIONAL SUMMARY

RTL Design Engineer (FPGA/ASIC) at XCLR Technologies, specializing in **RISC-V-based SoC** development. Certified **RISC-V Foundational Associate (RVFA)** and **Google Summer of Code (GSoC) 2022 contributor** and **2023 mentor**, with hands-on experience in **SystemVerilog/Verilog** based **ASIC** and **FPGA** designs, **Lint/CDC/RDC** sign-off, **timing closure**, and **verification**.

TECHNICAL SKILLS

- **RTL Design** — RISC-V | OOO Core | VLSU | JTAG | SerDes | PCIe | eFuse | UART | CDC | XBAR | Cache | AXI | APB | CHI | TileLink
- **EDA Tools** — VCS | Verdi | Xilinx Vivado | GTKWave | CoreAssembler | CoreConsultant | QuestaSim | Verilator | FusionCompiler | PrimeTime
- **Formal Verification** — JasperGold | SVA (SystemVerilog Assertions)
- **Static Analysis** — Synopsys SpyGlass (Lint,CDC,RDC)
- **Languages** — Verilog | SystemVerilog | SystemVerilog-OOP | UVM (basic) | C | C++ | RISC-V Assembly
- **Scripting** — Tcl | Shell | Makefile | CI/CD | Python | YAML (FuseSoC)
- **Version Control** — GitHub | GitLab
- **Operating System** — Linux | Windows

WORK EXPERIENCE

02/2025 – Present

RTL Design Engineer, *XCLR Technologies (Pvt.) Ltd.*

- **AWS EC2 F1 Shell-Compatible SoC Subsystem —** Developed and validated a subsystem making the SoC's **JTAG**, **UART**, and **DDR4** interfaces compatible with AWS EC2 F1 Shell's memory-mapped interface **(OCL/BAR1)** and **DDR** controller. Verified interface accessibility through **C++** host-side drivers across both F1 emulation and AWS EC2 simulation environments
- **VU19P FPGA SoC Bring-Up —** Led development and emulation of an SoC targeting the VU19P FPGA using **Xilinx Vivado** for **synthesis**, implementation, and **bitstream** generation. Integrated UART, DDR, and JTAG interfaces, and validated DDR and JTAG functionality through on-hardware emulation testing on the VU19P board
- **PULP BSCANE2 TAP Integration for Unified JTAG Access —** Integrated PULP's **dmi_bscane_tap** via **BSCANE2**, enabling a single **HS-3 cable** on the VU19P's JTAG TAP (XDC Pins) to handle both bitstream configuration and RISC-V debug access. Validated the path on-hardware using **OpenOCD/GDB**, testing **halt/resume**, **single-stepping**, **breakpoints**, and **register/memory access**
- **DDR4 Bring-Up & Linux Boot on AWS EC2 F1 FPGA —** Interfaced the SoC with AWS EC2 F1's **DDR4-C** shell controller to bring up external memory, then loaded the Linux boot image into DDR and validated end-to-end boot on the Ariane **(CVA6)** RISC-V core, confirming memory-mapped access and core-to-DDR data integrity through the shell interface
- **JTAG Bring-Up & Debug Validation on AWS EC2 F1 —** Bridged the design's JTAG interface to AWS EC2 F1 shell's JTAG connectivity, with **OpenOCD** running as the intermediate server translating RISC-V **GDB's** remote protocol (host-side) into JTAG scan commands driving the design's debug module (target-side) on the F1 emulation instance. Ran full test cycles covering core halt/resume, **single-stepping**, **breakpoints**, and **register/memory access**, validating consistent debug behavior across repeated test runs

09/2023 – 02/2025

Associate Engineer, *Xcelerium*

- **AWS EC2 F1 Shell Interface Validation —** Verified 4 AWS EC2 F1 shell interfaces **(OCL, BAR1, DDR, SDA)** using **C++ drivers** and SystemVerilog testbenches, across both AWS EC2 F1 emulation and simulation environments.
- **Logic-Analyzer-Based Debugging —** Debugged JTAG and SoC subsystem behavior using logic analyzers across both AWS Cloud emulation and VU19P FPGA hardware environments
- **Technical Documentation & Design Reviews —** Authored comprehensive design documents, verification test plans, timing diagrams, and block diagrams to aid design reviews and knowledge transfer

03/2020 – 09/2023

Research Assistant, *Micro Electronics Research Lab (MERL)*

- **RV32I Core Design —** Built an open-source RV32I CPU core from scratch in SystemVerilog, simulated with **Verilator** and **FuseSoC**, and emulated on **an Arty A7 FPGA** with **Xilinx Vivado**. Later integrated basic **UVM** and **JasperGold** support
- **Out-of-Order Upgrade to RV32I Core —** Added out-of-order execution: **2-wide fetch**, **RAT**-based register renaming, **ROB** for in-order commit, and reservation stations feeding **ALU/branch units** migrating the **pipeline** to a **dispatch/issue/commit** structure, **BOOM-inspired**
- **NOVA1 SoC Development —** Contributed to **NOVA1 System-on-Chip (SoC)** development and its **AWS EC2-based** emulation, and **Zyper OS** bring-up

- **Multicore Cache Coherency & CHI Research —** Led the research of the **multicore architecture** and its **cache coherency** through the **OpenPiton** framework. Also researched converting OpenPiton's native coherence protocol to ARM's **CHI**, mapping its directory-based coherence engine onto CHI's RN-**F/HN-F/SN-F** node abstractions
- **RISC-V Training & Mentorship —** Conducted **student training programs** on RTL design and how to contribute to open-source
- **NOVA1 Presentation —** Presented NOVA1 project at the **RISC-V Karachi Meetup**

SOFT SKILLS

Languages — English (C1), Urdu (Native)

Other — Design review | Technical Documentation | Debugging & Root-Cause Analysis | Design verification coordination | Technical Leadership | Timing-Closure Problem Solving | Research & Mentorship | Volunteer work

PROJECTS

Google Summer of Code (GSoC), *Contributor (2022) / Mentor (2023)*

- Added **FuseSoC** support on the **Azadi-SoC** and **rebased** the **Ibex core** to latest upstream commits (GSoC'22)
- Mentored **TileLink** Uncached Heavyweight **(TL-UH)** implementation in **Azadi-SoC** (GSoC'23)

MARCore & NOVA1, *Research Lead / Contributor*

- Researched multicore cache-coherent memory systems using OpenPiton workflow and Ariane Core (MARCore1)
- Contributed to NOVA1 SoC that contains RV64GC Core, UART, BRAM, DDR, and JTAG support (The NOVA Project)

EDUCATION

2019 – 2023

Bachelor of Engineering in Electrical Engineering, *Usman Institute of Technology*

(CGPA: 3.289) (Specialization: Computer Systems)

- **Major Courses** — Digital Logic Design | Computer Architecture | Microprocessor-Based Systems | Embedded Systems | Operating Systems

Honors & Awards

- Hasham Foundation Scholarships by Usman Institute of Technology in 2022
- Linux Foundation (LiFT) Scholarship Recipient in 2024

CERTIFICATES

RISC-V Foundational Associate (RVFA) — Date: Aug 2025 | Issuer: RISC-V International | Sponsor: Linux Foundation | Link

RISC-V Fundamental (LFD210) — Date: Aug 2025 | Issuer & Sponsor: Linux Foundation | Link

PUBLICATIONS & ARTICLES

- Multicore SoC For SMP Linux (Thesis)
- NOVA Emulation Project- An open-source RISC-V-based application-class AI SoC (Project Doc)
- A Summer of Opportunity (Medium — GSoC experience)
- Abstractions Can Lead Humanity to Ignorance (An Unpopular Opinion) (Medium — technical opinion)